

Aliah University

Department of Electrical Engineering, B-Tech

End-Semester Examination, Spring-2020

Subject: Basic Electrical Engineering [EENUGES01]

Full Marks: 40

Year: I

Time: 2 Hours

Semester: II

1. a) Distinguish between mesh and loop of a network. 2x5=10 Marks
- b) What is the utility of superposition theorem?
- c) Is Ohm's law true for all conductors? Name two types of commercially available resistors.
- d) Give two basic speed control scheme of DC shunt motor?
- e) Define power factor. State the conditions under which it is (i) maximum and (ii) minimum. #
- f) Can DC be applied to transformers?

Answer any three questions

3x10=30 Marks

2. a) Discuss the significance of power triangle. 2x5=10 Marks
- b) Derive the emf equation of a single phase transformer.
3. a) A 250 V DC shunt motor takes 40 A at full load. The resistances of the armature and shunt field windings are 0.5 ohm and 250 ohm, respectively. Determine the back emf on full load. 5 Marks
- b) A 50kVA, 5000/500V, 50 Hz, single phase transformer has the high and low voltage winding resistance of 8 ohm and 0.04 ohm, respectively. The no-load loss of the transformer is given as 900 W. Find the efficiency of the transformer at full load and 0.8 power factor lagging. 5 Marks
4. a) Derive the condition for obtaining maximum power from a source to resistive load. 2x5=10 Marks
- b) What is an ideal transformer? Draw phasor diagram at different loads for an ideal transformer.
5. a) For the circuit shown in **Fig. 1** obtain all nodal voltages. 2x5=10 Marks
- b) Determine current across AB (**Fig.2**) using Thevenin's theorem.

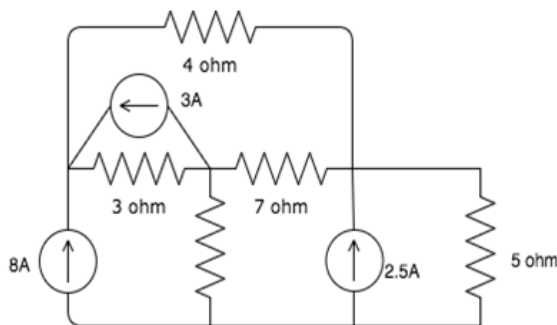


Fig.1

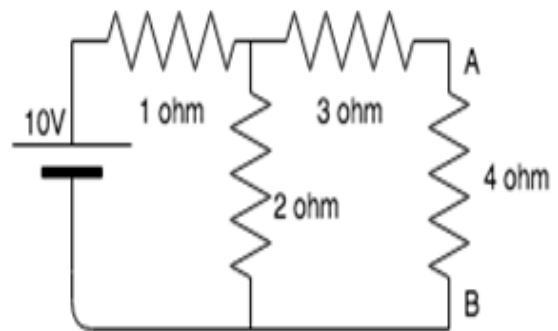


Fig. 2

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